Solar Performance Analyzer

Technical Report & Algorithm Design

Project Team

Ujjawal Verma
Khushboo Khilnaney
Satwik Srijan

Track: Himshikhar Data Science Track

July 14, 2026

Contents

1 Executive Summary & Problem Statement

The global transition to renewable energy heavily relies on the operational efficiency of solar power plants. However, a significant challenge in managing these assets is the prevalence of **invisible output gaps**. Solar plants frequently suffer from silent underperformance caused by factors such as panel fouling (dust and debris accumulation), thermal derating, and localized inverter faults. Because these issues do not always trigger hard offline alerts, they often go entirely unnoticed when observing aggregate generation data.

These undetected faults lead to compounding energy losses over time, severely reducing the asset's Return on Investment (ROI) and overall grid reliability. The **Solar Performance Analyzer** aims to resolve this critical blind spot. The proposed solution is an automated, machine-learning-driven analyzer designed to translate raw weather and generation data into highly accurate expected generation curves and actionable alerts, ultimately recovering lost revenue.

2 Data Architecture & Preprocessing

2.1 Data Sources and Pipeline

To accurately model solar output, the analyzer relies on high-fidelity telemetry data heavily integrated with strictly monitored weather inputs. The core variables ingested into the localized data warehouse include:

- **AC Power Output:** The actual generation from the inverters.
- **Solar Irradiation:** The primary driver of photovoltaic generation.
- **Module Temperature:** Panel surface temperatures impacting efficiency.
- **Ambient Temperature:** Atmospheric conditions surrounding the arrays.

Time-series data is continuously captured and streamed at strict 15-minute intervals. This high-resolution granularity provides necessary visibility into transient anomalies that might otherwise be smoothed out in hourly or daily aggregates.

2.2 Data Cleaning & Feature Engineering

Raw sensory data is inherently noisy. To ensure the machine learning model focuses purely on daylight generation constraints, the pipeline applies strategic removal of nighttime noise (specifically, filtering out periods where irradiation < 0.02). Advanced imputation algorithms handle missing values, preventing gap-induced model instability.

Furthermore, the data processing engine performs automated capacity normalization and feature engineering, which includes incorporating rolling averages and time-of-day offsets to capture diurnal generation patterns effectively.

2.3 Metric Standardization: Performance Ratio

Solar plants utilize inverters of vastly different scales and capacities. To evaluate large and small inverters on an identical scale, the system adopts the industry-standard **Performance Ratio (PR)**. This step transforms an absolute kW prediction into a dimensionless efficiency problem (ranging roughly from 0 to 1), effectively neutralizing scale biases.

The actual operating capacity of an inverter is estimated aggressively as the 95th percentile of AC power during optimal high-irradiance readings:

$$\text{Cap}_{\text{inv}} = Q_{0.95}\left(\text{Power}_t \mid \text{Irr}_t > 0.75\right) \tag{1}$$

Using this inferred capacity, the Performance Ratio for a given inverter i is calculated as:

$$\text{PR}_i = \frac{\text{Power}_i}{\text{Cap}_{\text{inv}}} \tag{2}$$

3 Machine Learning Framework

3.1 The Ensemble Model: Random Forest Regressor

The core predictor of the Solar Performance Analyzer is a **Random Forest Regressor**. This ensemble of decision trees is intentionally selected to minimize variance and aggressively prevent overfitting, which is common in highly volatile weather datasets.

By utilizing bootstrap-resampled data and selecting random feature subsets at every node split, the forest maintains robustness against noise. Each individual tree recursively splits the feature space to minimize the Mean Squared Error (MSE) within the resulting leaves.

The ensemble average is defined as:

$$\hat{y}(x) = \frac{1}{T}\sum_{t=1}^{T} f_t(x) \tag{3}$$

Where T is the total number of trees, and $f_t(x)$ is the prediction of the t-th tree. The decision splits optimize for maximum variance reduction (Information Gain):

$$\text{Gain} = \text{Var}(y_p) - \left(\frac{n_L}{n}\text{Var}(y_L) + \frac{n_R}{n}\text{Var}(y_R)\right) \tag{4}$$

3.2 Contamination Bias Fix: Robust Regression Trim

Standard regression models minimize errors over all provided data. However, if faulty operational periods are included in the training set, they pull the fitted curve downward, causing leverage bias. This results in an artificially lowered expectation of what "normal" generation looks like.

To counter this, the analyzer deploys a two-pass "healthy data" trim—a highly effective robust regression technique conceptually similar to RANSAC.

Pass 1 (Residual Calculation): An initial model fits the data, and residuals are calculated:

$$r_i = \frac{y_i - \hat{y}_i^{(1)}}{\hat{y}_i^{(1)}} \tag{5}$$

Pass 2 (Healthy Set Refit): The dataset is trimmed to exclude significantly under-performing data points, retaining only the "healthy" set:

$$\text{Healthy} = \{i : r_i > -0.15\} \tag{6}$$

Refitting the Random Forest solely on this subset forces the model to estimate what *should* happen during healthy states, rather than predicting the average operation inclusive of breakdowns.

4 Fault Diagnostics and Signal Processing

4.1 Hard-Threshold Classification

While the expected output is generated via machine learning, the diagnostic engine relies on deliberate, non-learned threshold logic to evaluate deviations. This crucial design decision ensures the system remains highly auditable and deterministic, generating statuses that operations teams can blindly trust without black-box opacity.

The performance gap is calculated as the percentage deviation from the expected output:

$$\text{Gap} = \frac{\text{Expected} - \text{Actual}}{\text{Expected}} \tag{7}$$

Based on the magnitude of the gap, the system triages the severity into actionable buckets:

$$\text{Status} = \begin{cases} \text{Critical} & \text{Gap} > 0.30 \\ \text{Warning} & 0.15 < \text{Gap} \le 0.30 \end{cases} \tag{8}$$

4.2 Nonparametric Anomaly Detection

At any given timestamp, all inverters within a specific solar plant theoretically share identical localized weather conditions. The analyzer utilizes peer benchmarking to compare inverters against one another.

Crucially, the system benchmarks using the **median** rather than the mean. If 2 out of 22 inverters are offline, a simple mean would drastically lower the expected levels for the entire plant. Using the median provides a robust central tendency that prevents failing hardware from masking adjacent failures. The peer deviation is computed as:

$$\mathrm{Dev}_i = \frac{\mathrm{median}(\mathrm{PR}_{\mathrm{peers}}) - \mathrm{PR}_i}{\mathrm{median}(\mathrm{PR}_{\mathrm{peers}})} \tag{9}$$

If the deviation metric breaches a 20% limit against healthy peers, it strongly reinforces the individual inverter fault alarms.

4.3 Run-Length Encoding and Persistence

Solar generation data is susceptible to transient noise, such as isolated passing clouds or temporary shading. To prevent alert fatigue, the analyzer employs Run-Length Encoding—a discrete-time signal processing technique functioning as a debouncing filter.

Alerts are physically triggered only when the negative run length extends to 4 or more consecutive 15-minute intervals (representing 1 full hour of sustained underperformance). State changes are tracked cleanly mathematically:

$$\mathrm{change}_t = \mathbb{I}[b_t \neq b_{t-1}] \tag{10}$$

5 Business Value & Explainability

5.1 Financial Impact Quantification

Alerts based purely on percentages lack business context. The analyzer transforms abstract performance ratios directly into actionable financial metrics using a discretized left-Riemann-sum over the 15-minute intervals.

$$\mathrm{Loss} = \max(\mathrm{Gap}, 0) \times \Delta t \tag{11}$$

By linearly scaling this lost kW volume against user-defined energy tariffs, asset managers can instantly view exact energy losses in monetary terms. This empowers teams to prioritize maintenance dispatches based on true financial value rather than arbitrary alert queues.

5.2 Explainable AI (XAI) and TreeSHAP

To ensure the machine learning predictions are interpretable, the platform leverages Explainable AI frameworks. Global importance metrics derived from Random Forest variance reduction reliably identify irradiation as the $\sim 99\%$ dominant predictive feature, providing a critical physics-based sanity check.

$$\mathrm{Imp}(j) = \frac{1}{T} \sum_t \sum_{\text{splits on } j} \Delta \mathrm{Var} \tag{12}$$

For local, granular explainability, the system utilizes Shapley Additive Explanations

(TreeSHAP). Shapley values decompose individual predictions into exact per-feature contributions, enabling precise root-cause diagnostics per alert.

$$\phi_j(x) = \sum_S \frac{|S|!(|F| - |S| - 1)!}{|F|!}[f(S \cup \{j\}) - f(S)] \tag{13}$$

6 Model Performance and Limitations

6.1 Performance Metrics

The diagnostic engine is rigorously evaluated against multiple dimensions to ensure reliability in production environments.

Metric	Definition	Importance for Operations
Accuracy	Overall correctness of standard operation	Baselines core model reliability during healthy daylight generation.
Precision	$\frac{\text{True Faults}}{\text{True Faults+False Alarms}}$	Maximizes trust and aggressively prevents alert fatigue for operators.
Recall	$\frac{\text{True Faults}}{\text{Total Actual Faults}}$	Ensures severe and critical plant failures are practically never missed.
RMSE	Root Mean Square Error	Measures exact kW deviation limits for the continuous regression baseline.

Table 1: Key Model Performance Metrics

6.2 Limitations and Responsible Use

While powerful, the Solar Performance Analyzer has inherent limitations that mandate responsible operational use:

- **Hardcoded Thresholds:** The persistence interval (4 intervals), gap thresholds (15%/30%), peer deviation limits (20%), and overheat cutoffs (55°C) are intentional judgment calls. They prioritize auditable business logic over complex statistical fits but may require regional tuning.
- **Data Dependency:** Regression predictions rely entirely on localized telemetry. Anomalies such as pyranometer sensor drift or physical misalignment can artificially mimic widespread inverter hardware failure.
- **Human-in-the-Loop:** The diagnostic engine is ultimately a high-level decision-support system. While it dictates financial prioritization and highlights anomalies, physical interventions and hardware replacements still strictly require manual validation by trained site engineers.

7 Conclusion and Future Goals

By fusing robust machine learning regression, localized explainability, and deterministic signal processing, the Solar Performance Analyzer successfully translates raw solar noise into actionable financial intelligence. It removes the ambiguity of hidden output gaps, transforming silent underperformance into visible, quantifiable maintenance priorities.

Future Goals: The next iteration of the pipeline aims to integrate dynamic 1-minute streaming telemetry for even faster transient detection, alongside live grid-pricing APIs to provide dynamic financial impact estimations based on real-time market rates.